

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture CFG - 1



Context Free Grammar (CFG)

Consider the regular expression $a(a^* \cup b^*)a$

Any string in this language consists of a leading a, followed by a middle part generated by $(a^* \cup b^*)$, followed by a trailing b.

If we let S be a new symbol interpreted as *a string in the language*

and M be a symbol standing for *middle part*

Then we can express this observation by writing

$S \rightarrow aMb$ where $\rightarrow$ is read *can be*.

We call such an expression a *rule*.

Context Free Grammar (CFG)

What can M , the middle part, be?

The answer is: either a string of a's or a string of b's.

We express this by adding the rules $M \rightarrow A$ and $M \rightarrow B$ where A and B are new symbols that stand for strings of a's and b's, respectively.

What is a string of a's?

It can be the empty string

$A \rightarrow e,$

OR it may consist of a leading a followed by a string of a's:

$A \rightarrow aA.$

Context Free Grammar (CFG)

Similarly, for B: $B \rightarrow e$ and $B \rightarrow bB$.

The language alternatively can be defined by the following language generator:

$S \rightarrow aMb$

$M \rightarrow A, M \rightarrow B$

$A \rightarrow e, A \rightarrow aA.$

$B \rightarrow e, B \rightarrow bB.$

Note

The class of languages that have some sort of *recursive structure*.

The definition is *non-deterministic*

Context free grammar

A *context free grammar* (*cfd*) G is a quadruple (V, Σ, R, S) where

V is an (finite) alphabet set

$\Sigma \subseteq V$ is the set of *terminals*

S (the *start symbol*) is an element of $V - \Sigma$

R (the *set of rules*) is a (finite) subset of $(V - \Sigma) \times \Sigma^*$

The members of $V - \Sigma$ are called *non-terminals*

Notation:

Whenever $(A, u) \in R$ we write $A \rightarrow_G u$

Context free language

For any $u, v \in \Sigma^*$, we write $u \Rightarrow_G v$ if and only if there are strings $x, y \in V^*$ & $A \in V - \Sigma$ such that $u = xAy$, $v = xv'y$ and $A \rightarrow_G v'$

The relation $\Rightarrow_G^*$ is the reflexive transitive closure of $\Rightarrow_G$

$L(G)$ the language generated by G is $\{w \in \Sigma^* : S \Rightarrow_G^* w\}$

A language L is said to be a *context free language* (*cfl*) if $L = L(G)$ for some *context free grammar* G .

We call any sequence of the form

$$w_0 \Rightarrow_G w_1 \Rightarrow_G \dots \Rightarrow_G w_n$$

a derivation in G of w_n from w_0 , where $w_i \in V^*$

We also say that the derivation has n steps

Context free grammar

Example: $L = \{a^n b^n : n \geq 0\}$

Consider $G = (V, \Sigma, R, S)$ where

$V = \{S, a, b\}$

$\Sigma = \{a, b\}$

R consists of the rules $S \rightarrow e$, $S \rightarrow aSb$

Consider a^2b^2 , then the possible derivation is

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$

The **1st two steps** used the rule $S \rightarrow aSb$

and the **last step** used the rule $S \rightarrow e$

We also write $S \rightarrow e \mid aSb$

CFL

Example

Language of all strings of properly balanced left and right parentheses, like $()(())$

We let $G = (V, \Sigma, R, S)$

where $V = \{S, (,)\}$, $\Sigma = \{ (,) \}$

R consists of the rules

$S \rightarrow e, S \rightarrow SS, S \rightarrow (S)$

Derivation of $()(())$

$S \Rightarrow SS \Rightarrow S(S) \Rightarrow S((S)) \Rightarrow S(()) \Rightarrow (S)(()) \Rightarrow ()(())$